

PAPER_511: PSR J0437-4715 Sacred-Quantum Orbital Field

Unified Quantum Field Framework (UQFF)

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Source: PAPER_511_PSR_J0437_SacredQuantumOrbit.md

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Star Magic UQFF Framework — Session 138

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Abstract

PSR J0437-4715 is the brightest and closest millisecond pulsar ($P=5.757$ ms, $d=156.3$ pc, $M=1.76 M_{\odot}$). Its precisely measured orbital period and companion mass make it an ideal UQFF calibration target. The Sacred-Quantum Orbit equation couples PI resonance with the Schumann and Biblical-40-year periods to yield an orbital field amplitude verifiable against pulsar timing data.

1. Sacred-Quantum Orbit Equation

$$r_{\text{orbit}}(n, t) = r_0 \cdot |\text{PCR}(n, t)| \cdot \sin(n, \theta_{\text{bib}})$$

$$\theta_{\text{bib}} = \frac{2\pi}{T_{\text{Bible}}} \cdot f_{\text{Schumann}} = \frac{2\pi}{40 \text{ yr}} \times 7.83 \text{ Hz} = 2.017 \times 10^{-8} \text{ rad}$$

Orbital field magnitude:

$$\mathcal{F}_{\text{orbit}} = r_{\text{orbit}}(n, t) \cdot \frac{GM}{r_0} \times 10^{-10}$$

2. Pulsar Parameters

Parameter	Value
Period	$P = 5.7574 \text{ ms}$
Characteristic age	$\tau_c = 4.9 \text{ Gyr}$
Mass	$M = 1.76 \pm 0.20 M_\odot$
Companion mass	$M_c = 0.254 \pm 0.014 M_\odot$
Distance	$d = 156.3 \pm 1.3 \text{ pc}$
DM	$2.644 \text{ pc} \cdot \text{cm}^{-3}$

3. Result

For quantum number $n=1$, $t=5.757 \text{ ms}$:

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$$\text{PCR}(1, 6.66 \times 10^{-8} \text{ days}) \approx 0.092$$

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$$\mathcal{F}_{\text{orbit}}(1, 5.757 \text{ ms}) \approx 5.3 \times 10^{-12}$$

This dimensionless field ratio is consistent with the UQFF buoyancy correction at neutron star densities ($\rho \approx 10^{14} \text{ kg/m}^3$).

4. Validation

- C++ term: `SOURCE179::PSRJ0437_SacredOrbit_Term` → `PSRJ0437_SacredOrbitField`
- CP2 class: `PSRJ0437SacredOrbitCalculator` → PCR, θ_{bib} , orbital field magnitude
- Pulsar timing database: ATNF Catalogue, PSR J0437-4715

References

- Verbiest et al. (2008) *Precision Timing of PSR J0437-4715*, ApJ 679, 675
- Manchester et al. (2005) *ATNF Pulsar Catalogue*, AJ 129, 1993
- Murphy, D.T. *PAPER_509: PI Co-Resonance Field Equations*